Rotaract Club of Xiamen University Malaysia | Rotary Club of Kuala Lumpur Dynamics | Rotary Club of Tun Razak Exchange

SDC Hackathon

Technical Tools Declaration

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Declare every API, cloud service, AI tool, dataset, model, and sponsor resource your team used to build the project. Complete every section; where a category does not apply, tick "No" or write "None" – do not leave it blank. Submit as a single PDF document named "SDC2026_TeamName_TechDeclaration.pdf" together with your project submission. Undeclared third-party resources discovered at judging may affect your score or eligibility.

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SDC HACKATHON 2026

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1. Team & Project Overview

Team Name	Debug The World
Project / Solution Name	ForgeFlow
Name (Leader)	ZHOU HONGRUI
Email & Phone (Leader)	+60 1117563302 era2609074@xmu.edu.my
University Affiliation(s)	Xiamen University Malaysia
Live Prototype URL	https://efc961db-e94c-4415-a08b-a75381586db6-00-158x0gilza3ss.pike.replit.dev/forgeflow/
Code Repository URL (public / access granted to judges)	https://github.com/RockPu/ForgeFlow-Hackathon

2. Architecture & Technology Stack

2.1 System Architecture Overview

Describe how the components declared in this document connect end-to-end (e.g., user interface → backend → AI model/API → database → cloud hosting). Attach a system block diagram as an appendix if available.

[Maximum 200 words]

ForgeFlow is a role-based factory order and product traceability web app. The React/TypeScript/Vite frontend uses a same-origin /api interface to a Node.js/Express API. The API validates inputs with Zod, enforces role permissions, and reads or writes PostgreSQL through Drizzle ORM using DATABASE_URL. The database stores demo users, sessions, orders, inventory, production and inspection events, dispatch, and audit records. The prototype is hosted on Replit and can also run locally with PostgreSQL. Its risk panel uses deterministic fictional examples; the application does not call an AI model or external business-risk API.

2.2 Languages, Frameworks & Libraries

List the main programming languages, frameworks, and third-party libraries that your team did not create. Open-source use is permitted provided it is declared and properly licensed/attributed.

Name & Version	Purpose in Project	License
TypeScript ~5.9.3	Frontend and API language	Apache-2.0
React 19.1.0	Web user interface	MIT
Vite ^7.3.2	Frontend development and build	MIT
Node.js >=22.12; Express ^5.2.1	Server runtime and REST API	MIT (Node.js / Express)
PostgreSQL 16+; Drizzle ORM ^0.45.2	Relational database and ORM	PostgreSQL License; MIT
TanStack Query ^5.90.21; Zod ^3.25.76	API state management and schema validation	MIT

3. APIs & Cloud Services

3.1 External APIs

List every external API your solution calls (GenAI, maps, payments, messaging, data feeds, etc.). Do not include API keys or secrets in this document.

API & Provider	Purpose in Project	Access Tier (free / paid / sponsor-provided)
None - no external business APIs used	No AI, credit, maps, ERP, payment, messaging, or carrier API	N/A

3.2 Cloud Platforms & Deployment

List every cloud service used for hosting, storage, databases, authentication, or compute, and indicate how the prototype is deployed.

Service & Provider	Purpose in Project	Plan / Region
Replit	Project workspace and hosted prototype preview	Hackathon Replit Core; region not stated
PostgreSQL 16+	Persistent relational data via DATABASE_URL	Provider / region not specified in exported source

4. AI Tools, Models & Datasets

4.1 AI Models Used in the Solution

List every AI model embedded in or called by your application, including pre-trained, fine-tuned, and self-trained models.

Model & Version	Source / Provider	Usage (pre-trained / fine-tuned / self-trained)	License / Terms
None - no AI model is embedded in or called by the application	N/A	N/A	N/A

4.2 Datasets

List all datasets used for training, fine-tuning, evaluation, or in the running application. State clearly if a dataset contains personal or sensitive data.

Dataset Name	Source / URL	Type (public / private / self-collected / synthetic)	License	Personal Data? (Y/N)
ForgeFlow seed / demo fixtures	Project source: artifacts/api-server/src/factory/fixtures.ts	Synthetic; project-created	Project-created; no external dataset	N

4.3 AI-Assisted Development Tools

Tools used to help write code, design, content, or documentation (e.g., chat assistants, code completion tools, image generators). Declaring these tools does not reduce your score; undeclared use may.

Did your team use any generative-AI tool to help create code, design, text, or other project material?

☑ Yes ☐ No

If yes, list them below:

Tool & Version	**Used For**	**Extent of Use (minor / moderate / major)**
Replit Agent	Code generation, implementation, and debugging	Major

Briefly describe how your team reviewed, tested, and took responsibility for AI-generated output:

[Maximum 100 words]

AI-generated code and text were reviewed by team members against the project requirements and revised during development. The team retains responsibility for implementation choices and submitted materials. Prototype workflows were manually reviewed; AI output was not treated as an independent security or correctness audit.

5. Sponsor-Provided Resources

Declare any official SDC sponsor-provided GenAI API, cloud credits, tools, mentoring, or data that your team used.

Did your project use any official SDC sponsor-provided resource?

☑ Yes ☐ No

If yes, list them below:

Sponsor	Resource (API / credits / platform / data)	How It Was Used in the Project
SDC Hackathon	Replit Core / Replit platform	Required development workspace, AI-assisted coding, and hosted prototype preview

Was the sponsor resource essential to the core functionality of the prototype, or could it be replaced? Explain briefly:

[Maximum 100 words]

Replit supported the required hackathon development workflow and prototype preview. The application does not depend on a proprietary Replit API; its source can run on compatible Node.js and PostgreSQL hosting. Replit was useful for collaboration and judging access, but another compatible host could replace it for deployment.

6. Security, Privacy & Responsible Use

6.1 Data Handling & Privacy

What user or personal data does your application collect, store, or send to third-party services (including AI APIs)? How is it protected, and how long is it kept? State "No user data is collected" if applicable.

[Maximum 150 words]

This prototype uses fictional demo companies, orders, inventory, production events, and tracking data; do not enter real personal, customer, or financial information. Demo accounts store account identifiers and roles; passwords are scrypt-hashed. Session tokens are hashed in storage, cookies are HttpOnly and SameSite=Strict (Secure in production), and sessions expire after 8 hours. Role checks and CSRF protection are used. Application records persist in the configured PostgreSQL database until reset or deleted. No data is sent to external AI or credit APIs. A production retention policy is not defined.

6.2 Known Limitations & Risks

Identify the main technical limitations of your AI components (e.g., hallucination, bias, rate limits, dependency on a single provider) and any mitigation you have implemented.

[Maximum 150 words]

The risk panel displays deterministic fictional sample assessments; it performs no live company research, credit checks, or AI inference. Sample evidence is unverified and must not guide real credit decisions; the human Manager makes the final approval decision. The prototype lacks ERP, procurement, payment, carrier-label, and notification integrations and is not hardened for production. Demo credentials are public and must never be reused. Use synthetic data and add production security, privacy, backups, and monitoring before real deployment.

7. Compliance & Readiness Checklist

Tick all boxes that apply before submitting.

☑ All APIs, cloud services, AI tools, models, and datasets used in the project are declared in this document.

☑ All third-party code, datasets, and models are used in accordance with their licenses, and attribution is provided where required.

☑ No API keys, passwords, or other credentials are exposed in the code repository, prototype, or this document.

☑ Any use of generative-AI tools for development has been disclosed and reviewed by the team.

☑ Any use of official SDC sponsor resources has been declared in Section 5.

☑ The prototype described here is accessible to judges through the links provided in Section 1.

☑ All fields in this template have been completed within the word limits stated.

8. Declaration

By submitting this document, the team leader confirms that the information provided is complete and accurate to the best of the team's knowledge, and that every API, cloud service, AI tool, dataset, model, and sponsor resource used in the project has been declared.

[Signature] ZHOU HONG RUI

[Date] 23/10/2026